

Unit 12, Lesson 10: Evaluating Real-World Problems Using the Order of Operations – Part 2

Benchmark

MA.B.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers, including exponents and radicals.

Learning Target

- ☐ **I can solve real-world problems involving the order of operations with rational numbers, including exponents and radicals.**

12.10.1 Warm-Up: Bell Work

Find the value of each expression. Show your work.

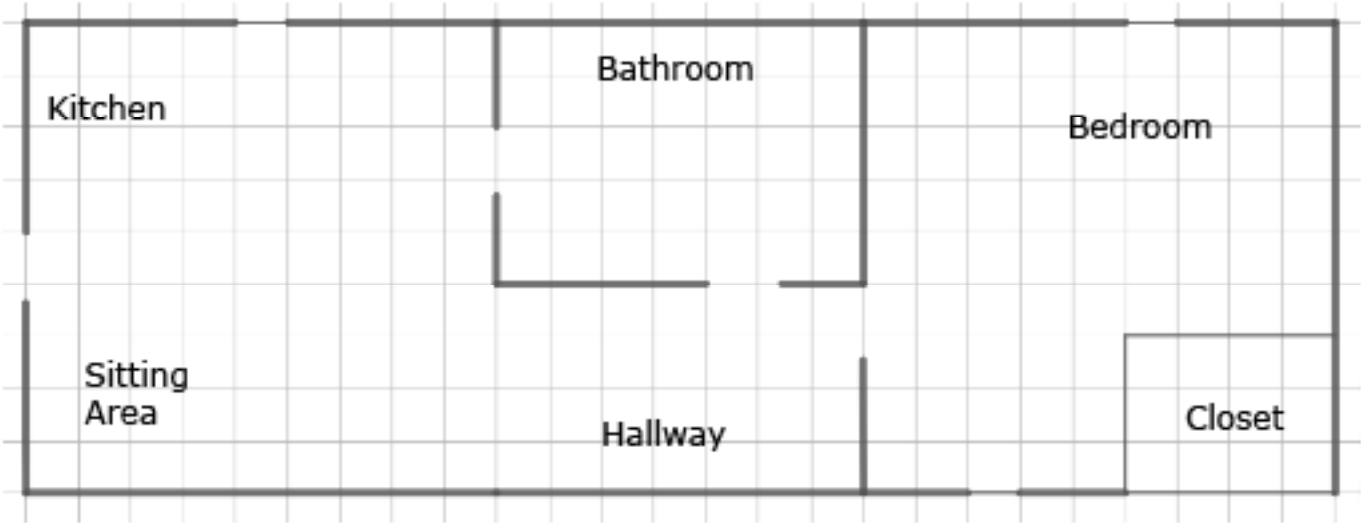
1. $\left(3 \cdot \frac{2}{3}\right)^{-2} - 4\left(6 + \frac{7}{2}\right)$

2. $\frac{55^8 - 925}{7} - \sqrt{11 - 2}$

12.10.2 Activity: Station 1: Home Renovations

Liliana is looking to do some home renovations on her apartment. The floor plan of her apartment is shown, where each gridline measures 1 foot (ft). She plans to re-tile the floor of the bathroom and lay new carpeting on the floor of the bedroom, including the closet.

Her bedroom, including the closet, measures 9 ft by 9 ft, and her bathroom is 5 ft by 7 ft.



The prices Liliana found for the materials she needs to update the flooring in both rooms is given in the table.

	Bedroom			Bathroom	
	Carpet	Padding Material	Underlay	Tile	Grout
	Home Decorators Sycamore	Memory Foam	OSA	Marazzi	Polyblend
Price	\$2.89/ft ²	\$0.75/ft ²	\$1.00/ft ²	\$1.89/ft ²	\$13.48

1. Lilitana writes the following expression to determine the total cost of installing flooring in both rooms.

$$9^2(2.89 + 0.75 + 1) + (5 \cdot 7)(1.89) + 13.48$$

- Discuss with your partner how the expression relates to the materials in the table.
- Circle the part of the expression that represents the cost of carpeting the bedroom.
- Underline the part of the expression that represents the cost of tiling the bathroom.
- Evaluate the expression to determine how much it will cost Lilitana for the materials to replace the flooring in both rooms. Show your work.

$$9^2(2.89 + 0.75 + 1) + (5 \cdot 7)(1.89) + 13.48$$

- Lilitana is concerned about the total cost and wants to know how much it would cost to re-floor the bedroom only. How much would the flooring materials cost for only her bedroom?

12.10.3 Activity: Station 2: Generating Power

1. After several power outages caused by storms, Phillip purchased a used generator for \$1,380 three years ago. Equipment loses value each year from basic wear and tear and as newer models come on the market. Phillip estimates the generator loses about 20% of its value each year based on prices he sees online.
 - a. Phillip uses the expression $1380 \cdot 0.8^3$ to determine a fair price for which to sell the generator now. What is the fair price Phillip determined?
 - b. If Phillip's estimation of how much value the generator loses each year is accurate, he can use the expression $1380 \cdot 0.8^{-2}$ to determine the price of the generator two years before he bought it when it was new. Evaluate the expression to find the price of the generator when it was new.

12.10.4 Activity: Station 3: Planting Strawberries

1. **Michaela is considering planting strawberries for the first time on her farm. She has an acre set aside and wants to determine whether the costs and potential profits make sense.**

After researching, she determines the following information.

Costs of Planting 1 Acre of Strawberries	Expected Revenue from 1 Acre of Strawberries
- Plants: \$1500 - Labor: \$595 - Materials: \$1724	- Total plants: 15,000 - Expected percentage available for sale: 70% - Average revenue per plant: \$1.21

The expression shown can be used to find the amount of money Michaela can expect to earn based on her research.

$$1.21(0.70 \cdot 15,000) - (1500 + 595 + 1724)$$

- a. **Discuss with your partner how the expression relates to the information in the table.**

- b. Profit can be calculated using the equation shown.**

$$\text{Revenue} - \text{Costs} = \text{Profit}$$

Work with your partner to explain how this relates to the expression Michaela wrote.

- c. Determine the profit Michaela is expecting to earn based on her research.**

12.10.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each learning target in the table.

 I need help.

 I got it.

 I could help someone.

Evaluate expressions with rational numbers, exponents, and radicals using the order of operations.	  
Solve real-world problems by evaluating expressions using the order of operations.	  